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CMPG321

Advanced Databases

Business Intelligence and
Data Warehouses

SU4 – Ch13 (13th) or Ch15 (3rd)

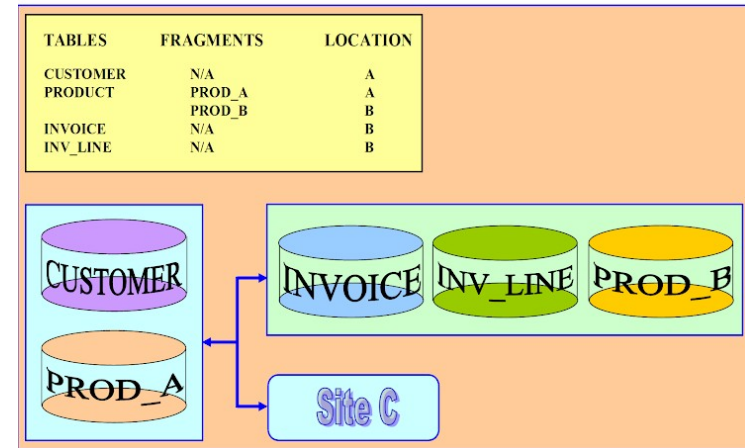
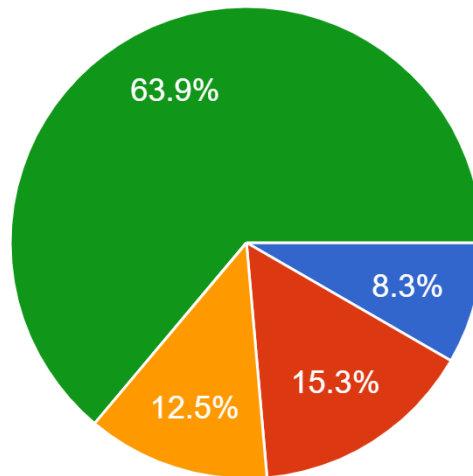
Monday 12 August

#DiscoverNWU



Attendance of 05 August

```
BEGIN;  
  
INSERT CUSTOMER(CUS_NUM, CUS_NAME, CUS_ADDRESS, CUS_BAL)  
  
VALUES ('34210','Victor Ephanor', '123 Main St', 0.00);  
  
INSERT INTO INVOICE(INV_NUM, CUS_NUM, INV_DATE, INV_TOTAL)  
  
VALUES ('986434', '34210', '10-AUG-2018', 2.00);  
  
COMMIT;
```



- Remote request
- Remote transaction
- Distributed request
- Distributed transaction

SU4 - Outcomes

Business Intelligence and Data Warehouses

- Describe how business intelligence provides a comprehensive business decision support framework
- Discuss the business intelligence architecture, reporting styles, evolution and benefits
- Differentiate between operational data and decision support data
- Identify the purpose, characteristics, and components of a data warehouse
- Develop star and snowflake schemas for decision-making purposes
- Describe the characteristics and capabilities of online analytical processing (OLAP)
- Explain the role and functions of data analytics, and the use of SQL analytic functions



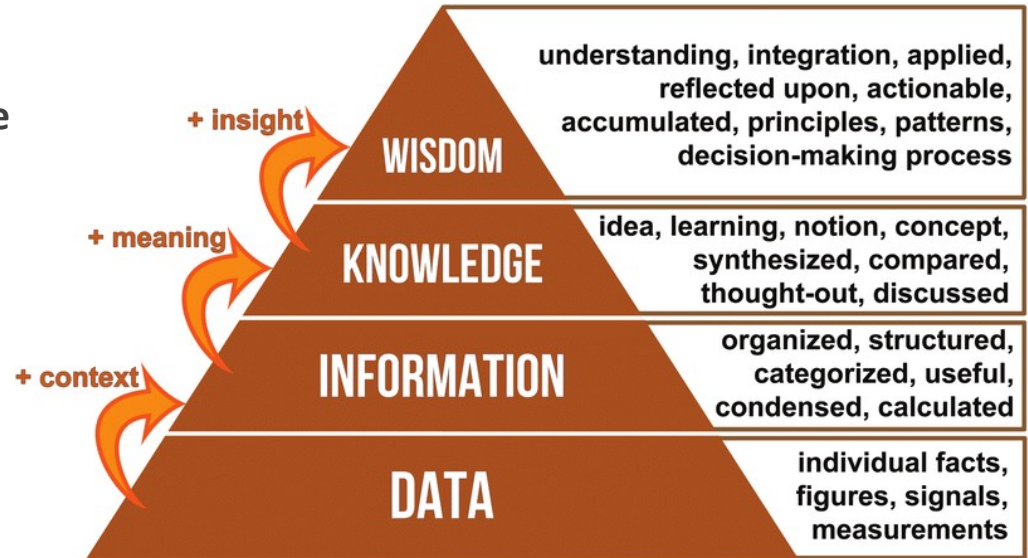
The Need for Data Analysis...

- Organizations tend to grow and prosper as they gain a better understanding of their environment
 - Evaluate through tracking daily transactions and analyzing company data
- Organizations are always looking for a competitive advantage
 - Product development, market positioning, sales promotions, and customer service
- Companies and software vendors addressed these multilevel decision support needs by creating autonomous applications for particular groups of users
 - This more comprehensive and integrated decision support framework within organizations became known as business intelligence



Business Intelligence (1 of 2)

- Comprehensive, cohesive, integrated set of tools and processes
 - Captures, collects, integrates, stores, and analyzes data
 - Generates and presents information to support business decision making
- Allows transformation
 - **Data** into **information**
 - **Information** into **knowledge**
 - **Knowledge** into **wisdom**





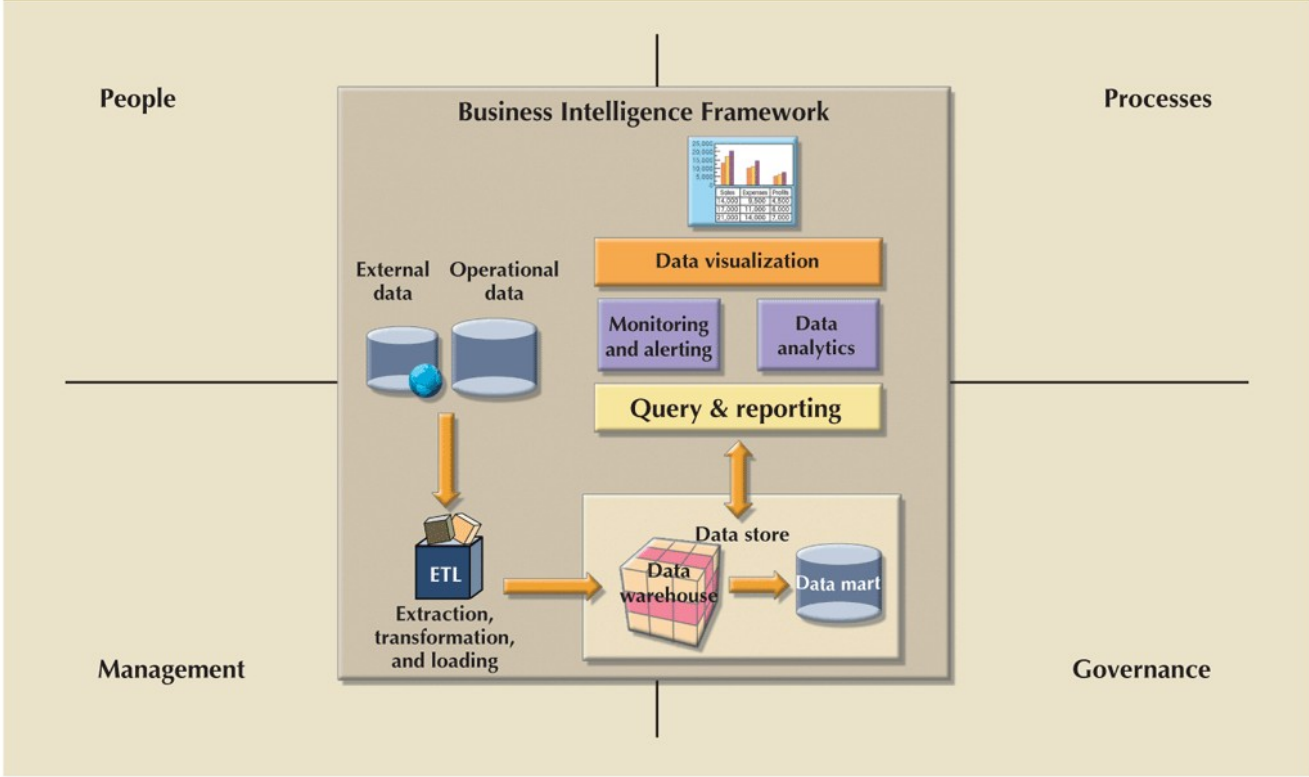
Business Intelligence (2 of 2)

- Concepts, practices, tools and techniques to help business
 - Understand core capabilities
 - Provide snapshots of the company situation
 - Identify key opportunities to create a competitive advantage
- Provides a **framework**
 - Collecting and storing operational data and aggregating it into decision support data
 - Analyzing decision support data and presenting generated information to end users to **support business decisions**
 - Making business decisions which generate more data
 - Monitoring results to evaluate outcomes and predicting future outcomes with a high degree of accuracy



Business Intelligence Architecture (1 of 3)

FIGURE 13.1 BUSINESS INTELLIGENCE FRAMEWORK





Business Intelligence Architecture (2 of 3)

Table 13.2 Basic BI Architectural Components	
Component	Description
ETL tools	Data extraction, transformation, and loading (ETL) tools collect, filter, integrate, and aggregate internal and external data to be saved into a data store optimized for decision support.
Data store	The data store is optimized for decision support and is generally represented by a data warehouse or a data mart. The data is stored in structures that are optimized for data analysis and query speed.
Query and reporting	This component performs data selection and retrieval, and it is used by the data analyst to create queries that access the database and create the required reports.
Data visualization	This component presents data to the end user in a variety of meaningful and innovative ways. This tool helps the end user select the most appropriate presentation format, such as summary reports, maps, pie or bar graphs, mixed graphs, and static or interactive dashboards.
Data monitoring and alerting	This component allows real-time monitoring of business activities. The BI system will present concise information in a single integrated view. This integrated view could include specific metrics about the system performance or activities, such as number of orders placed in the last four hours, number of customer complaints by product by month, and total revenue by region. Alerts can be placed on a given metric; once the value of a metric goes below or above a certain baseline, the system will perform a given action, such as emailing shop floor managers, presenting visual alerts, or starting an application.
Data analytics	This component performs data analysis and data-mining tasks using the data in the data store. This tool advises the user as to which data analysis tool to select and how to build a reliable business data model. Business models are generated by special algorithms that identify and enhance the understanding of business situations and problems.



Business Intelligence Architecture (3 of 3)

- Practices to manage data
 - **Master data management (MDM):** collection of concepts, techniques, and processes for identification, definition, and management of data elements (what business entity is of interest, what do we need from it, how do we obtain it)
 - **Governance:** method of government for controlling business health and for consistent decision making (a system of rules, practices and processes by which a company is directed and controlled)
 - **Key performance indicators (KPI):** numeric or scale-based measurements that assess company's effectiveness in reaching its goals (e.g. increasing sales per employee by 10%)
 - General
 - Finance
 - Human resources
 - Education
- Modern BI reporting styles
 - Advanced reporting
 - Monitoring and alerting
 - Advanced data analytics



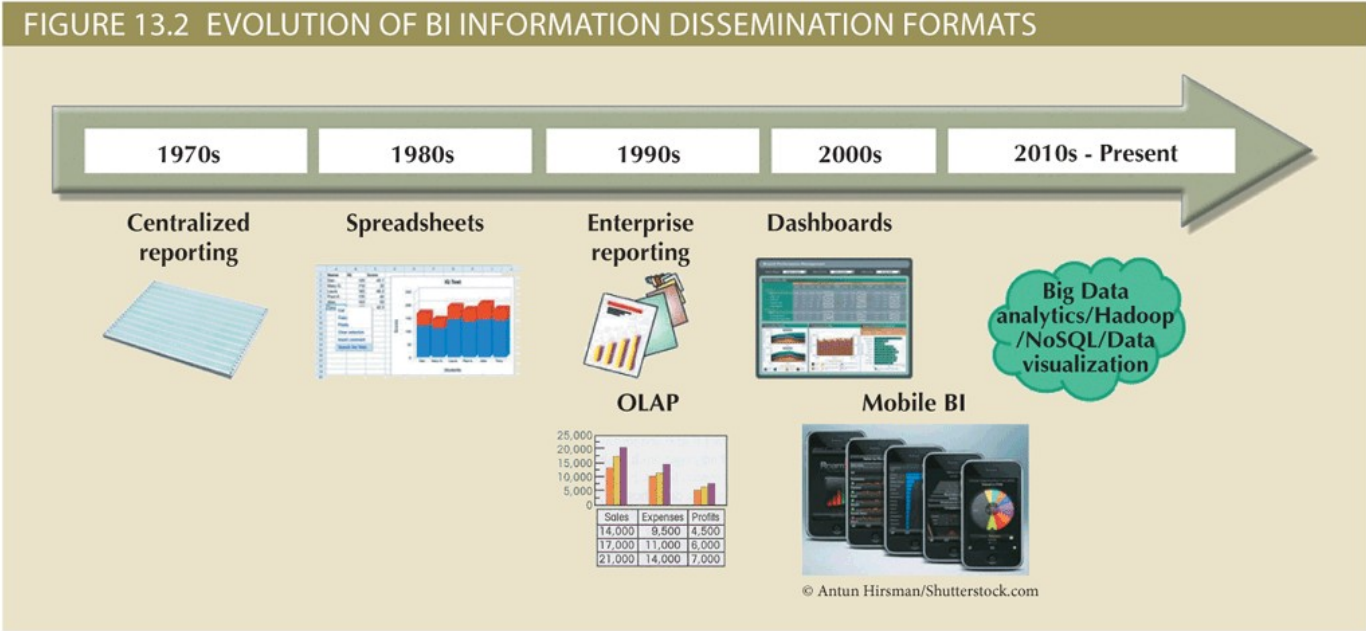
Business Intelligence **Benefits**

- **Improved decision making is the main goal** of BI, but BI provides other benefits
 - Integrating architecture (from all data sources)
 - Common user interface for data reporting and analysis
 - Common data repository fosters single version of company data
 - Improved organizational performance
- Achieving all these benefits takes a lot of human, financial, technological resources, and time
 - BI benefits are not achieved overnight; are the result of a focused company-wide effort that could take a long time





Business Intelligence Evolution (1 of 2)





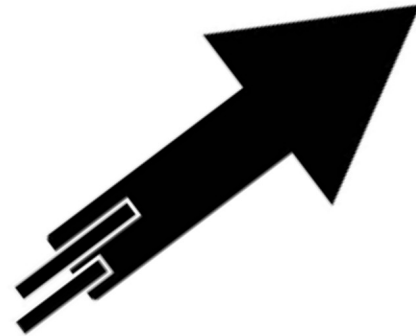
Business Intelligence Evolution (2 of 2)

- **Decision support system (DSS)** is an arrangement of computerized tools used to assist managerial decision making
 - Typically has a much narrower focus and reach than a BI solution
- BI information technology has evolved from centralized reporting styles to the current, mobile BI and Big Data analytics style in the span of just a few years
 - The rate of technological change is not slowing down; technology advancements are accelerating the adoption of BI to new levels



Business Intelligence Technology Trends

- Several technological advances are driving the growth of business intelligence technologies
 - Data storage improvements
 - Business intelligence appliances
 - Business intelligence as a service
 - Big Data analytics
 - Personal analytics





Decision Support Data

- Although BI is used at the strategic and tactical managerial levels within organizations, its effectiveness depends on the quality of data gathered at the operational level
 - Operational data in its basic form is seldom well suited to decision support tasks



Operational Data versus Decision Support Data (1 of 3)

- Operational data and decision support data serve different purposes
 - **Operational data** is useful for capturing daily business transactions
 - **Decision support data** gives tactical and strategic business meaning to the operational data
- Decision support data differs from operational data in three main areas (how it is used..)
 - **Time span**
 - **Granularity (level of aggregation)**
 - **Drill down:** Decomposing data into more atomic components (to a lower level)
 - **Roll up:** Aggregating data into summarized components (to a higher level)
 - **Dimensionality**



Operational Data versus Decision Support Data (2 of 3)

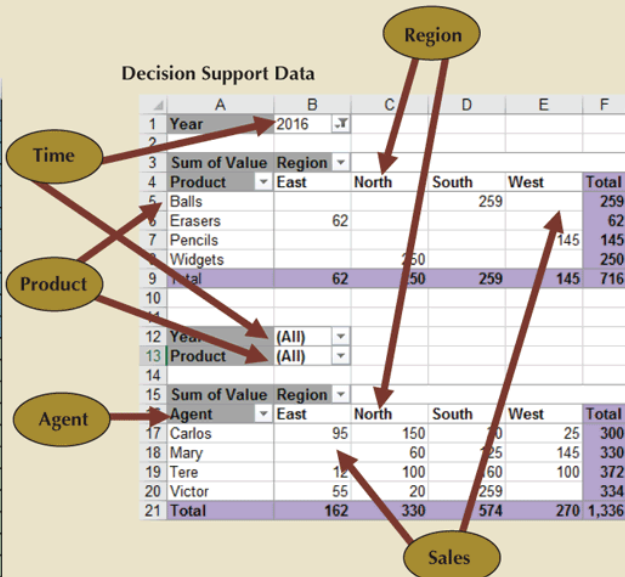
FIGURE 13.3 TRANSFORMING OPERATIONAL DATA INTO DECISION SUPPORT DATA

Operational Data

	A	B	C	D	E
1	Year	Region	Agent	Product	Value
2	2016	East	Carlos	Erasers	50
3	2016	East	Tere	Erasers	12
4	2016	North	Carlos	Widgets	120
5	2016	North	Tere	Widgets	100
6	2016	North	Carlos	Widgets	30
7	2016	South	Victor	Balls	145
8	2016	South	Victor	Balls	34
9	2016	South	Victor	Balls	80
10	2016	West	Mary	Pencils	89
11	2016	West	Mary	Pencils	56
12	2017	East	Carlos	Pencils	45
13	2017	East	Victor	Balls	55
14	2017	North	Mary	Pencils	60
15	2017	North	Victor	Erasers	20
16	2017	South	Carlos	Widgets	30
17	2017	South	Mary	Widgets	75
18	2017	South	Mary	Widgets	50
19	2017	South	Tere	Balls	70
20	2017	South	Tere	Erasers	90
21	2017	West	Carlos	Widgets	25
22	2017	West	Tere	Balls	100

Operational data has a narrow time span, low granularity, and single focus. Such data is usually represented in tabular format, in which each row represents a single transaction. This format often makes it difficult to derive useful information.

Decision Support Data



Decision support system (DSS) data focuses on a broader time span, tends to have high levels of granularity, and can be examined in multiple dimensions. For example, note these possible aggregations:

- Sales by product, region, agent, and so on
- Sales for all years or only a few selected years
- Sales for all products or only a few selected products



Operational Data versus Decision Support Data (3 of 3)

Table 13.5 Contrasting Operational and Decision Support Data <u>Characteristics</u>		
Characteristic	Operational Data	Decision Support Data
Data currency	Current operations Real-time data	Historic data Snapshot of company data Time component (week/month/year)
Granularity	Atomic-detailed data	Summarized data
Summarization level	Low; some aggregate yields	High; many aggregation levels
Data model	Highly normalized Mostly relational DBMSs	Non-normalized Complex structures Some relational, but mostly multidimensional DBMSs
Transaction type	Mostly updates	Mostly query
Transaction volumes	High-update volumes	Periodic loads and summary calculations
Transaction speed	Updates are critical	Retrievals are critical
Query activity	Low to medium	High
Query scope	Narrow range	Broad range
Query complexity	Simple to medium	Very complex
Data volumes	Hundreds of gigabytes	Terabytes to petabytes



Decision Support Database Requirements

- Database schema
 - Must support complex, non-normalized data representations
 - Data must be aggregated and summarized
 - Queries must be able to extract multidimensional time slices
- Data extraction and filtering
 - Allow batch and scheduled data extraction
 - Support different data sources and check for inconsistent data or data validation rules
 - Encourage advanced integration, aggregation, and classification
- Database size
 - Very large databases (VLDBs)
 - Advanced storage technologies
 - Multiple-processor technologies



The Data Warehouse (1 of 3)

- **Def:** “an integrated, subject-oriented, time-variant, nonvolatile collection of data that provides support for decision making”
- Collection of data
 - Provides support for decision making
- Components
 - Integrated
 - Subject-oriented
 - Time-variant
 - Nonvolatile



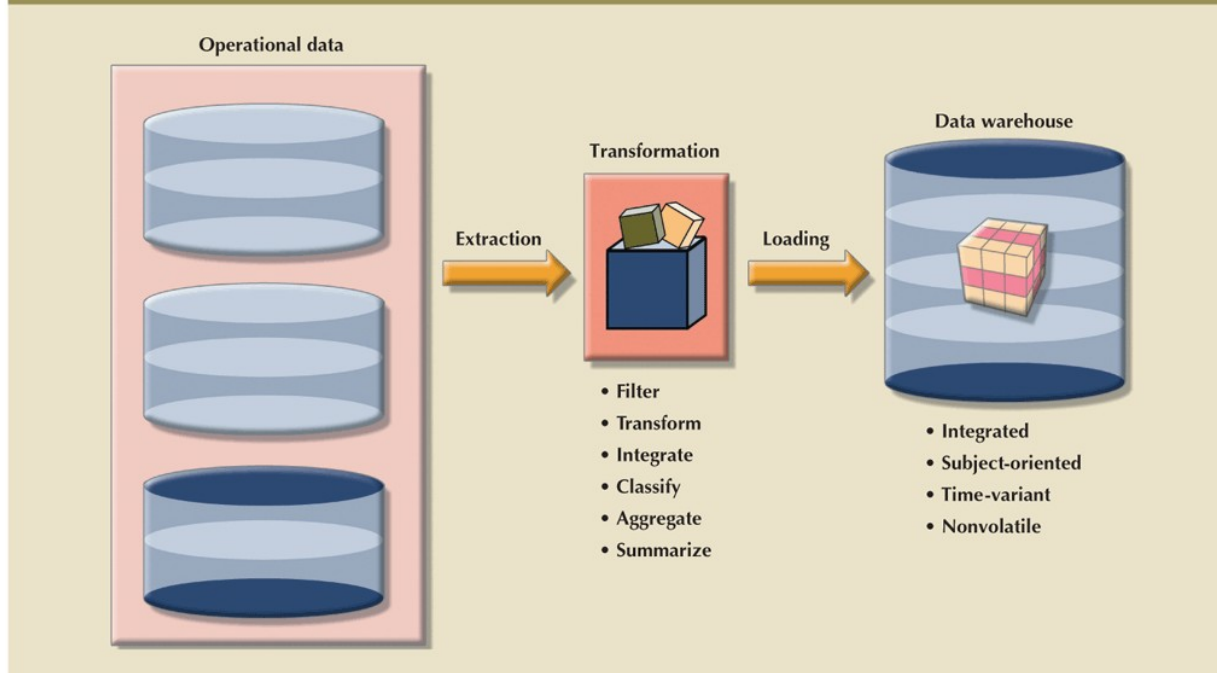
The Data Warehouse (2 of 3)

Table 13.8 Characteristics of Data Warehouse Data and Operational Database Data		
Characteristic	Operational Database Data	Data Warehouse Data
Integrated	Similar data can have different representations or meanings. For example, Social Security numbers may be stored as ###-##-#### or as #####, and a given condition may be labeled as T/F or 0/1 or Y/N. A sales value may be shown in thousands or in millions.	Provide a unified view of all data elements with a common definition and representation for all business units.
Subject-oriented	Data is stored with a functional, or process, orientation. For example, data may be stored for invoices, payments, and credit amounts.	Data is stored with a subject orientation that facilitates multiple views of the data and decision making. For example, sales may be recorded by product, division, manager, or region.
Time-variant	Data is recorded as current transactions. For example, the sales data may be the sale of a product on a given date, such as \$342.78 on 12-MAY-2016.	Data is recorded with a historical perspective in mind. Therefore, a time dimension is added to facilitate data analysis and various time comparisons.
Nonvolatile	Data updates are frequent and common. For example, an inventory amount changes with each sale. Therefore, the data environment is fluid.	Data cannot be changed. Data is added only periodically from historical systems. Once the data is properly stored, no changes are allowed. Therefore, the data environment is relatively static.



The Data Warehouse (3 of 3)

FIGURE 13.4 THE ETL PROCESS





Data Marts

- Small, single-subject data warehouse subset
 - Provides decision support to a small group of people
- Benefits over data warehouses
 - Lower cost and shorter implementation time
 - Technologically advanced
 - Inevitable “people issues”



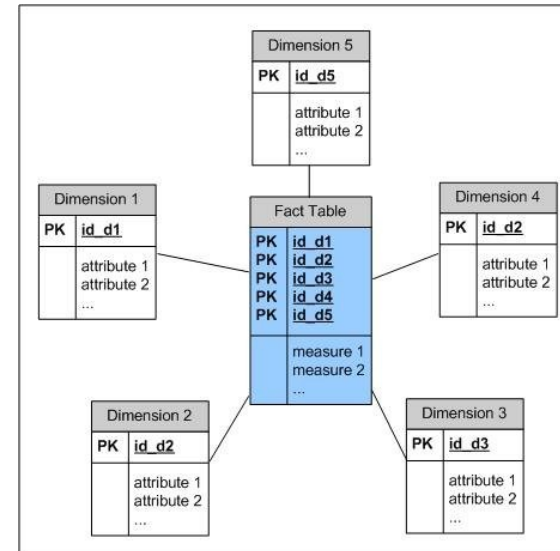
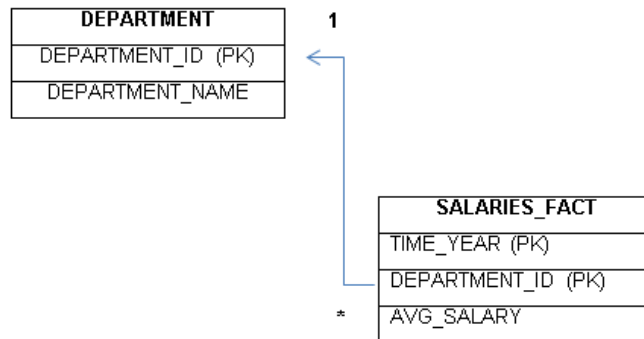
Twelve Rules That Define a Data Warehouse

Table 13.9	Twelve Rules for a Data Warehouse
Rule No.	Description
1	The data warehouse and operational environments are separated.
2	The data warehouse data is integrated.
3	The data warehouse contains historical data over a long time.
4	The data warehouse data is snapshot data captured at a given point in time.
5	The data warehouse data is subject oriented.
6	The data warehouse data is mainly read-only with periodic batch updates from operational data. No online updates are allowed.
7	The data warehouse development life cycle differs from classical systems development. Data warehouse development is data-driven; the classical approach is process-driven.
8	The data warehouse contains data with several levels of detail: current detail data, old detail data, lightly summarized data, and highly summarized data.
9	The data warehouse environment is characterized by read-only transactions to very large data sets. The operational environment is characterized by numerous update transactions to a few data entities at a time.
10	The data warehouse environment has a system that traces data sources, transformations, and storage.
11	The data warehouse's metadata is a critical component of this environment. The metadata identifies and defines all data elements. The metadata provides the source, transformation, integration, storage, usage, relationships, and history of each data element.
12	The data warehouse contains a chargeback mechanism for resource usage that enforces optimal use of the data by end users.



Star Schemas (1 of 5)

- Data-modeling technique
 - Maps multidimensional decision support data into a relational database
 - Creates the near equivalent of multidimensional database schema from existing relational database
 - Yields an easily implemented model for multidimensional data analysis
- Very simple Star Schema example





Star Schemas (2 of 5)

- Basic star schema components
 - **Facts:** numeric values that represent a specific business aspect
 - **Dimensions:** qualifying characteristics that provide additional perspectives to a given fact
 - **Attributes:** used to search, filter, and classify facts
 - Slice and dice: ability to focus on slices of the data cube for more detailed analysis
 - **Attribute hierarchies:** provide a top-down data organization
 - Aggregation and drill-down/roll-up data analysis



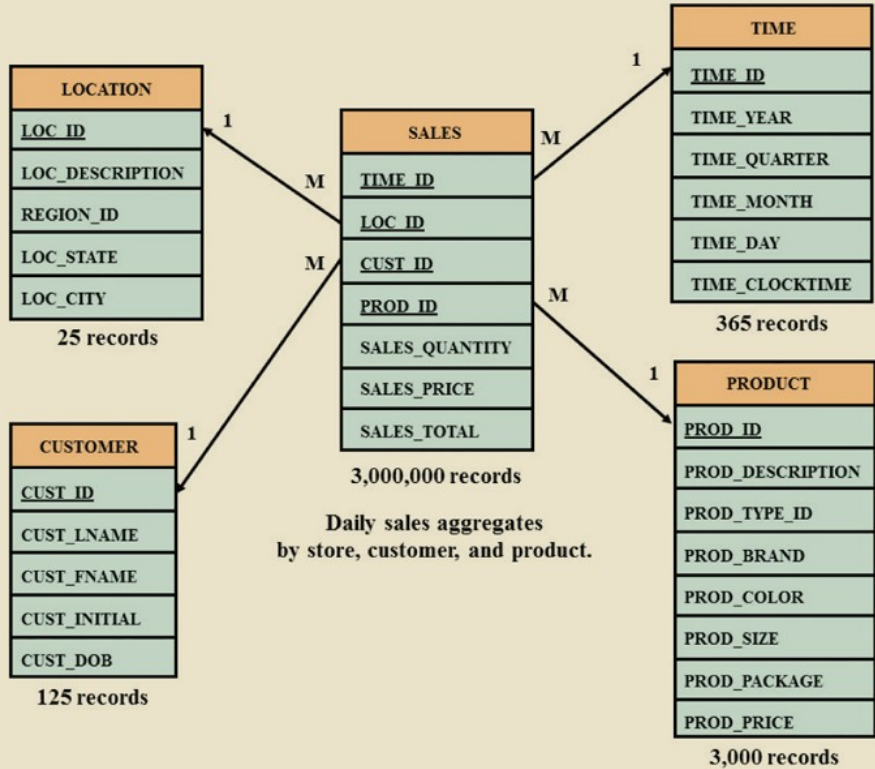
Star Schemas (3 of 5)

- Star schema representation
 - Facts and dimensions represented by physical tables in data warehouse database
 - Many-to-one (M:1) relationship between fact table and each dimension table
- Fact and dimension tables
 - Related by foreign keys
 - Subject to primary and foreign key constraints
 - Primary key of a fact table
 - **Composite primary key** because the fact table is related to many dimension tables
 - Always formed by combining the foreign keys pointing to the related dimension tables



Star Schemas (4 of 5)

FIGURE 13.10 STAR SCHEMA FOR SALES





Star Schemas (5 of 5)

- Performance-improving techniques for the star schema
 - **Normalizing dimensional tables** (Fig 13.12)
 - **Snowflake schema**: dimension tables can have their own dimension tables
 - **Maintaining multiple fact tables** to represent different aggregation levels (Fig 13.13)
 - Save processor cycles at run time, thereby speeding up data analysis
 - **Denormalizing fact tables**
 - Improves data access performance and saves data storage space
 - **Partitioning and replicating tables**
 - **Partitioning**: splits tables into subsets of rows or columns and places them close to the client computer
 - **Replication**: makes copy of table and places it in a different location
 - **Periodicity**: provides information about the time span of the data stored in the table



Online Analytical Processing (OLAP)

- Online analytical processing (OLAP) creates an advanced data analysis environment that supports decision making, business modelling, and operations research
- Online analytical processing (OLAP) is a BI style whose systems share three main characteristics
 - Multidimensional data analysis techniques
 - Advanced database support
 - Easy-to-use end-user interfaces





OLAP - Multidimensional Data Analysis Techniques

- Data are processed and viewed as part of a multidimensional structure
- Augmented advanced functions
 - Advanced Data presentation
 - Advanced Data aggregation, consolidation, and classification
 - Advanced Computational
 - Advanced Data-modeling



OLAP - Advanced Database Support

- OLAP tools must have the following features to deliver efficient decision support:
 - Access to many different kinds of DBMSs, flat files, and internal and external data sources
 - Access to aggregated data warehouse data and operational database detail data
 - Advanced data navigation features
 - Rapid and consistent query response times
 - Ability to map end-user requests
 - Support for very large databases



OLAP - Easy-to-Use End-User Interface

- Proper implementation leads to simple navigation and accelerated decision making or data analysis
 - Advanced OLAP features are more useful when access is kept simple
 - Many interface features are borrowed from previous generations of data analysis tools



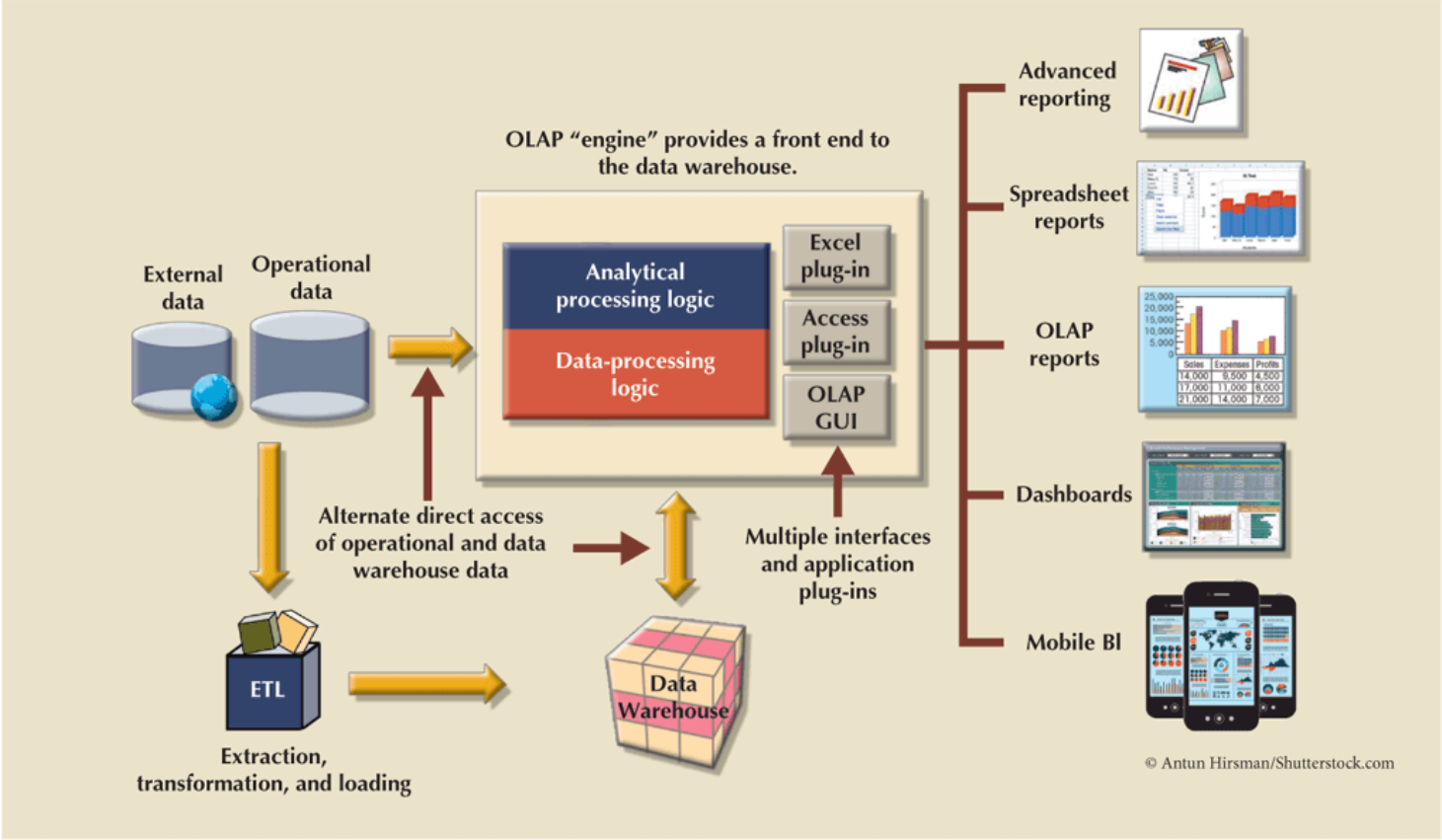
OLAP Architecture (1 of 3)

- Designed to meet ease-of-use requirements while keeping the system flexible
- Main architectural components
 - Graphical user interface (GUI)
 - Analytical processing logic
 - Data-processing logic



OLAP Architecture (2 of 3)

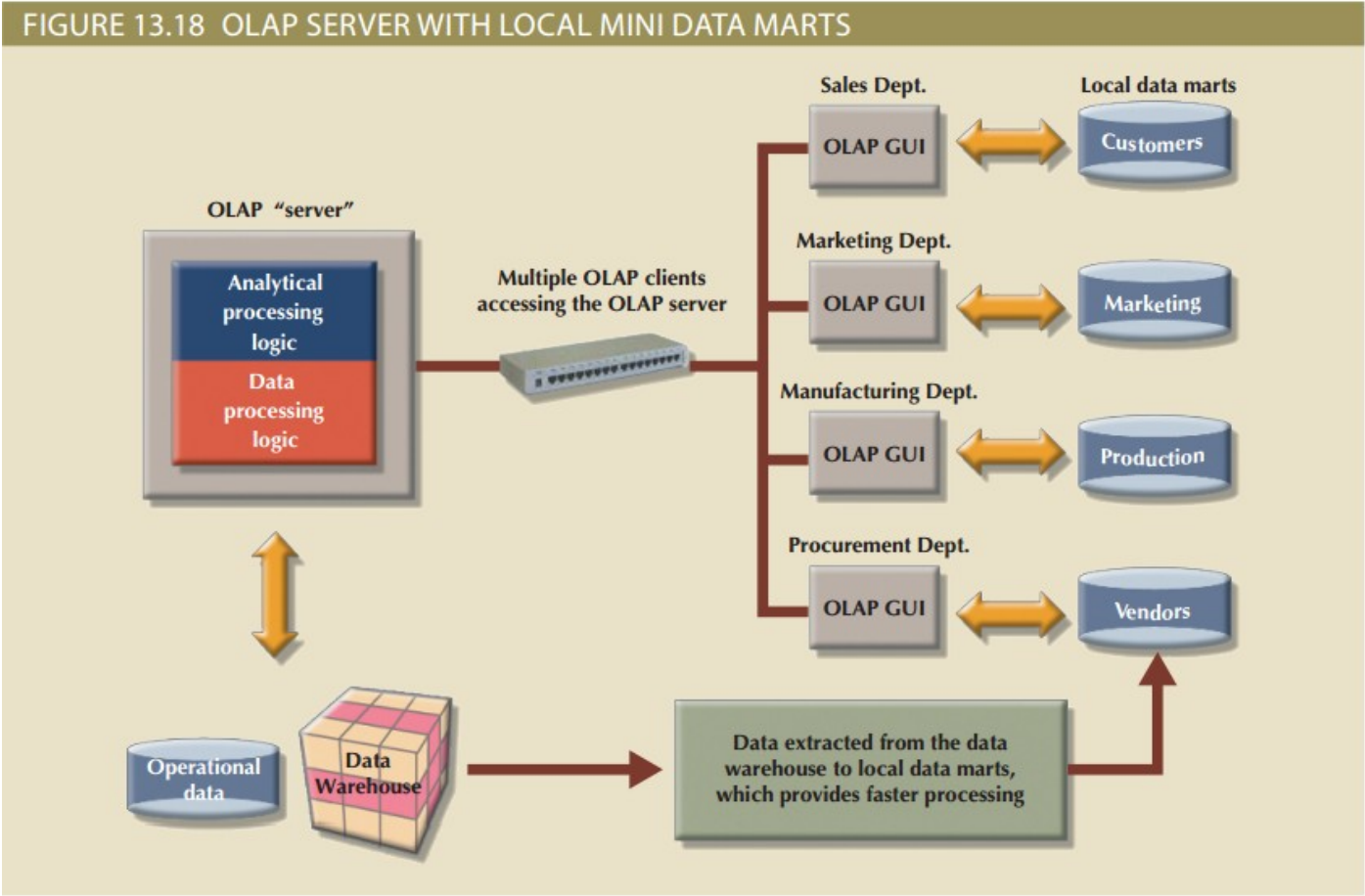
FIGURE 13.16 OLAP ARCHITECTURE





OLAP Architecture (3 of 3)

FIGURE 13.18 OLAP SERVER WITH LOCAL MINI DATA MARTS





Relational OLAP

- Relational online analytical processing (**ROLAP**)
 - Provides OLAP functionality using relational databases and familiar relational tools to store and analyze multidimensional data
- Extensions added to traditional RDBMS technology
 - Multidimensional data schema support within the RDBMS
 - Data access language and query performance optimized for multidimensional data
 - Support for very large databases (VLDBs)



Multidimensional OLAP

- Multidimensional online analytical processing (**MOLAP**)
 - Extends OLAP functionality to multidimensional database management systems (MDBMSs)
 - MDBMS uses proprietary techniques store data in matrix-like n -dimensional arrays
 - End users visualize stored data as a three dimensional **data cube**
 - Grow to n number dimensions, thus becoming hypercubes
 - Held in memory in a cube cache to speed access
 - **Sparsity**: measures density of data held in the data cube



Relational versus Multidimensional OLAP

Table 13.12 Relational vs. Multidimensional OLAP		
Characteristic	ROLAP	MOLAP
Schema	Uses star schema Additional dimensions can be added dynamically	Uses data cubes Multidimensional arrays, row stores, column stores Additional dimensions require re-creation of the data cube
Database size	Medium to large	Large
Architecture	Client/server Standards-based	Client/server Open or proprietary, depending on vendor
Access	Supports ad hoc requests Unlimited dimensions	Limited to predefined dimensions Proprietary access languages
Speed	Good with small data sets; average for medium-sized to large data sets	Faster for large data sets with predefined dimensions

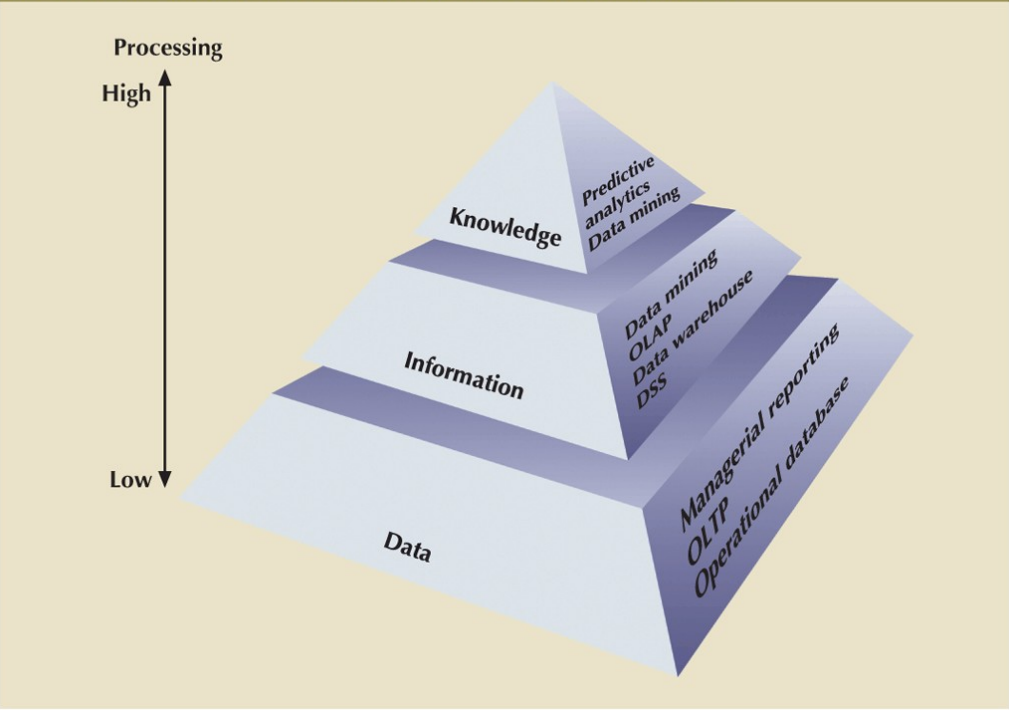


Data Analytics (1 of 4)

- Subset of business intelligence (BI) functionality that encompasses a wide range of mathematical, statistical, and modeling techniques with the purpose of extracting knowledge from data
 - Explanatory analytics: focuses on discovering and explaining data characteristics and relationships based on existing data
 - Predictive analytics: focuses on predicting future data outcomes with a high degree of accuracy
- Data mining focuses on the discovery and explanation stages of knowledge acquisition
 - Analyzing massive amounts of data to uncover hidden trends, patterns, and relationships; to form computer models to simulate and explain the findings; and to use such models to support business decision making



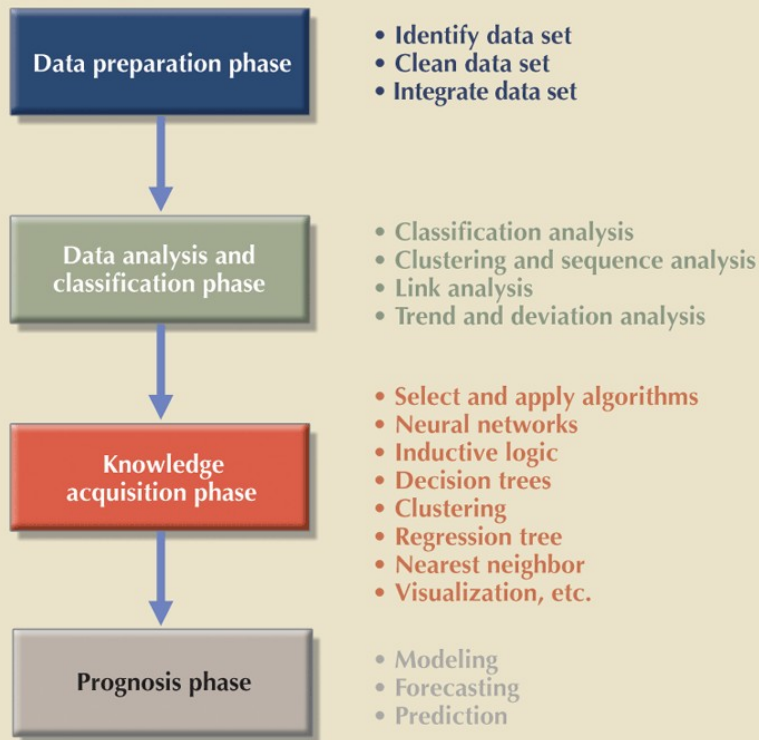
FIGURE 13.18 EXTRACTING KNOWLEDGE FROM DATA





Data Analytics (3 of 4)

FIGURE 13.19 DATA-MINING PHASES





Data Analytics (4 of 4)

- Predictive analytics focuses on creating actionable models to predict future behaviors and events
 - Employs mathematical and statistical algorithms, neural networks, artificial intelligence, and other advanced modeling tools to create actionable predictive models based on available data
 - Used in areas such as customer relationships, customer service, customer retention, fraud detection, targeted marketing, and optimized pricing



SQL Analytic Functions (1 of 2)

- The **ROLLUP** extension (Fig 13.21)
 - Used with GROUP BY clause to generate aggregates by different dimensions
 - Enables subtotal for each column listed except for the last one, which gets a grand total
- The **CUBE** extension (Fig 13.22)
 - Used with GROUP BY clause to generate aggregates by the listed columns
 - Enables you to get a subtotal for each column listed in the expression, in addition to a grand total for the last column listed



SQL Analytic Functions (2 of 2)

- Materialized views
 - Dynamic table that contains SQL query command to generate rows and stores the actual rows
 - Created the first time query is run and summary rows are stored in the table
 - Automatically updated when base tables are updated
 - Requires specified privileges



Data Visualization (1 of 2)

- Process of abstracting data to provide a visual data representation that enhances the user's ability to comprehend the meaning of the data
 - Goal is to allow the user to quickly and efficiently see the data's big picture by identifying trends, patterns, and relationships
- The need for data visualization
 - Ability to zoom in and out, drill down and up, filter, etc. is one of the many advantages of current breed of data visualization tools
 - Makes it easier to understand data—in particular, large amounts of data
- The science of data visualization
 - Roots in cognitive sciences: how the human brain receives, interprets, organizes, and processes information
 - Pattern recognition
 - Spatial awareness
 - Aesthetics



Data Visualization (2 of 2)

- Understanding the data
 - The same data can be presented in multiple ways
- In general, there are two types of data:
 - Qualitative: describes qualities of the data
 - Nominal or ordinal
 - Quantitative: describes numeric facts or measures of the data



Summary (1 of 2)

- Business intelligence (BI) is a term for a comprehensive, cohesive, and integrated set of applications used to capture, collect, integrate, store, and analyze data with the purpose of generating and presenting information to support business decision making
- Decision support systems (DSSs) refer to an arrangement of computerized tools used to assist managerial decision making within a business
- Operational data is not well suited for decision support
- The data warehouse is an integrated, subject-oriented, time-variant, nonvolatile collection of data that provides support for decision making
- The star schema is a data-modeling technique used to map multidimensional decision support data into a relational database for advanced data analysis



Summary (2 of 2)

- Online analytical processing (OLAP) refers to an advanced data analysis environment that supports decision making, business modeling, and operations research
- Data analytics is a subset of BI functionality that provides advanced data analysis tools to extract knowledge from business data
- Data mining automates the analysis of operational data to find previously unknown data characteristics, relationships, dependencies, and trends
- SQL has been enhanced with analytic functions that support OLAP-type processing and data generation
- Data visualization provides visual representations of data that enhance the user's ability to comprehend the meaning of the data

Huiswerk oefening / Homework exercise

Ch13 - Problem 2 – a,b,c (p. 649)

Victoria Ephanor manages a small product distribution company. Because the business is growing fast, Ephanor recognizes that it is time to manage the vast information pool to help guide the accelerating growth. Ephanor, who is familiar with spreadsheet software, currently employs a sales force of four people. She asks you to develop a data warehouse application prototype that will enable her to study sales figures by year, region, salesperson, and product. (This prototype is to be used as the basis for a future data warehouse database.)

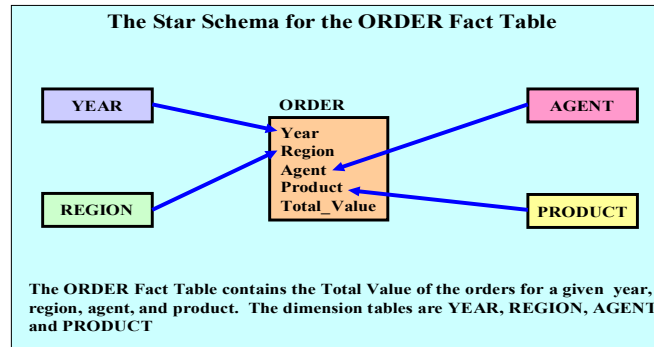
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2017	South	Mary	Widgets	75
2017	South	Mary	Widgets	50
2017	South	Tere	Balls	70
2017	South	Tere	Erasers	90
2017	West	Carlos	Widgets	25
2017	West	Tere	Balls	100

Problem 2 – a,b,c (p. 649)

- a. Identify the appropriate fact table components.**
- b. Identify the appropriate dimension tables.**
- c. Draw a star schema diagram for this data warehouse.**

Problem 2 – a,b,c (p. 649)

- Identify the appropriate fact table components.**
(Year, Region, Agent, Product)
- Identify the appropriate dimension tables.**
- Draw a star schema diagram for this data warehouse.**



CMPG321

- Marks updated on eFundi Gradebook (Attendance, Quizzes, Peer Assessments)
- *Test 1 - Marks not finalized yet....*
- Homework assignment
 - Revision Quiz SU3 – Closes Friday
 - Assignment 2 – Available Wednesday @09:30
 - Go through (Ch13 - 13th ed.; Ch15 – 3th ed.)
Business Intelligence and Data Warehouses
 - **Group Project**
 - Phase 1 - Due 21 August

